MUHAMAD FAHREZ BUKHORI

Electrical Engineering Undergraduate | IoT & Hardware | Control Systems | Automation

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PROFILE

Electrical Engineering undergraduate at UPN Veteran Jakarta with a vocational background in electrical engineering and hands-on experience in electrical maintenance, troubleshooting, and IoT hardware development. Active in KSM IoT's hardware engineering division; participated in FASTFEST UPH and FNC Bandung and received the Best Staff award. Interested in IoT, automation, CNC, PID control, and 3D printing.

EXPERIENCE

Hardware Engineering Staff | KSM Internet of Things, UPN Veteran Jakarta *Feb 2025 - Present*

- Contribute to hardware development and implementation for IoT and engineering projects.
- Collaborate with the team on troubleshooting and system development, applying problem-solving and teamwork skills.
- Participated in FASTFEST UPH and FNC Bandung and received the Best Staff award for contributions to the organization.

Member, Electronics and Control Research Team | UPN Veteran Jakarta *Oct 2026 - Present*

Content Developer | Sudut Pandang UPNVJ *May 2025 - Feb 2026*

Engineering Intern | Savero Hotel Depok *Jan 2023 - Jun 2023*

- Assisted in maintaining and inspecting electrical and electronic equipment throughout the hotel.
- Supported troubleshooting of electrical installations and control systems and performed routine facility inspections.
- Coordinated with relevant teams to address technical issues and followed occupational health and safety (OHS) procedures.

VOLUNTEER EXPERIENCE

Teaching Volunteer | TEMA 2025, HMTE UPNVJ / TBM Kolong *Sep 2025*

- Supported children in developing foundational reading and writing skills as part of the teaching team.
- Helped deliver interactive and supportive learning activities to build children's confidence and motivation.

EDUCATION

Universitas Pembangunan Nasional "Veteran" Jakarta (UPNVJ) | B.Eng. in Electrical Engineering *Expected Aug 2028*

Relevant studies: electrical power, electronics, control systems, IoT, hardware, automation, troubleshooting, and engineering system design.

SMK Negeri 2 Depok | Electrical Engineering *Graduated 2024*

Fundamentals of electrical installation, equipment maintenance, control systems, electrical measurement, troubleshooting, and workplace safety.

SKILLS

Technical: Electronics, Electrical Troubleshooting, Control Systems, IoT, PLC, Electrical Panels
Software & Programming: Python, R, Arduino IDE, AutoCAD, Autodesk Fusion 360
Areas of Interest: Automation, PID Control, CNC, 3D Printing
Languages: Indonesian, English

CERTIFICATIONS

Certified Associate in Health, Safety, and Environment (CAHSE) - ESAS Management, Mar 2025
KKNI Level 2 in Electrical Power Installation - BNSP (National Professional Certification Agency), Mar 2024 - Present
Certified competent in PLC programming and troubleshooting, electrical panel assembly and installation, and electrical circuit analysis and repair, with an emphasis on workplace safety.

PROJECTS

3 kg Sumo Robot

- Developed a 3 kg sumo robot using a BTS7980 motor driver for motor control. The robot is operated by remote control to perform movements and maneuvers in the arena.

Transporter Robot

- Developed a transporter robot to move objects by integrating the drive mechanism, electrical circuits, and motion control system, followed by functional testing.

Automated Guided Vehicle (AGV) - Ongoing

- Developing an AGV as an ongoing robotics research project. The project aims to establish a novel contribution and support a future manuscript submission to a journal indexed in SINTA or Scopus.